EQUIPMENT USED

TCT cutting tool – or beading & parting tool/HSS scraper
Centre drill
Thread-cutting lubricant
Parting tool
Hacksaw
Tap holder
Abranet and Micro-Mesh abrasives
Farecla polishing compounds

MATERIAL REQUIREMENTS

Ebonite, alumilite resin, aluminium or brass for the section
Nib unit, feed and ink converter

1 After an initial unsuccessful attempt using a polyester resin blank, which proved too brittle, I used some leftover aluminium rod to make the section. I mounted a short length of the rod in a scroll chuck fitted with small dovetail jaws

2 I used a tungsten-carbide tipped tool to cut a tenon for the 8.5mm section to barrel thread, measuring carefully to ensure the correct dimensions

3 The next step was to drill out the bore for the ink converter. This needs to be a good close fit so accurate drilling is of paramount importance. I used a centre drill to create a pilot hole, this helps to ensure that any subsequent drilling is centred and true

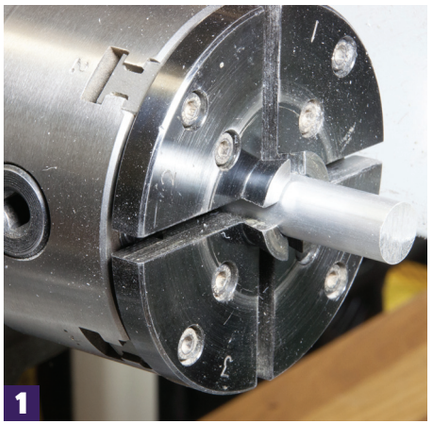
4 I then measured the diameter of the ink converter, mine was 6.35mm (¼in) so I drilled out to just under size at 6.3mm and then used an imperial ¼in drill to achieve an exact fit. I cannot stress enough that the fit of the converter is one of the keys to success in this project. A sloppy fit will result in disastrous ink leakages

5 Once the converter was a good fit I began the process of cutting the threads on the tenon. With the die mounted in the tailstock die holder and having applied a spray of thread-cutting lubricant, I turned the die and workpiece against one another by hand, slowly and carefully cutting and backing off to clear swarf

6 To finalise the thread up to the section body I reversed the die in the holder so the flat face was facing the work

7 Using a parting tool I then cut a recess at the rear of the thread so that when the body of the pen is attached it will fit snugly and flush to the section with no unsightly gaps

8 An important factor in achieving a good fit and a working section is careful measurement of the length required to fit the nib and feed. I assembled the components and measured with a steel rule. My measurements were accurate enough but Vernier callipers would perhaps be a better alternative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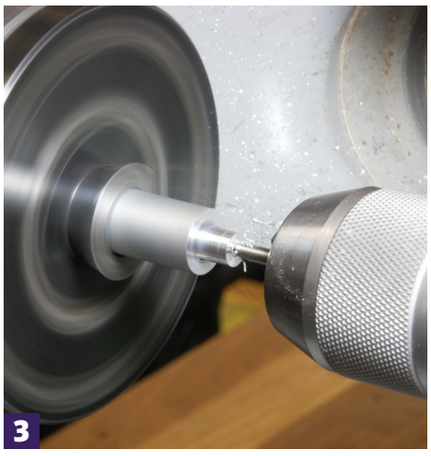
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